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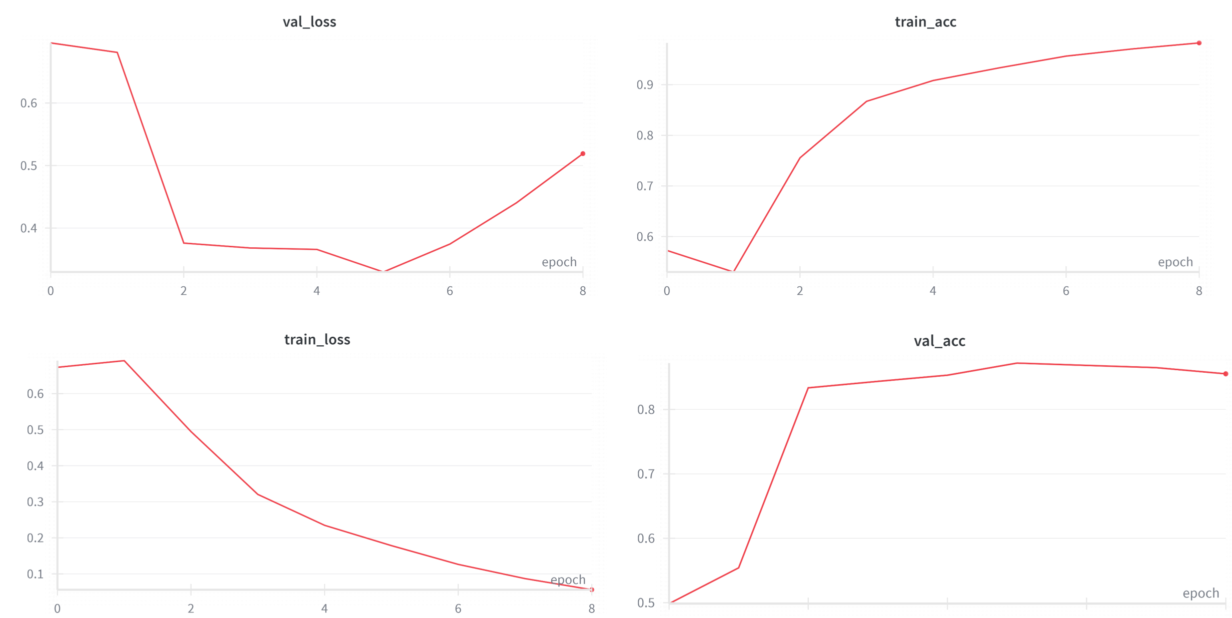
Assignment 3: IMDB Sentiment Classification using Bi-LSTM

Hyperparameters

Model architecture: Bidirectional LSTM
Max review length: 256. Reviews that are shorter are padded and reviews that are longer are shortened.
Embedding Dimension: 128. Size of the embedding for each token.
Hidden Dimension: 256. Size of the Bi-LSTM memory state.
Dropout: 0.3
Batch Size: 32
Optimizer: AdamW
Learning Rate: 0.001
Loss Function: Cross Entropy Loss
Early Stopping: Early stopping occurs after 3 epochs of no improvement in validation accuracy.
Random Seed: 42

Loss and accuracy curves from W&B

The model converged pretty quickly after about 3-4 epochs, with a steady decline in training loss. This shows the success of the Bi-LSTM architecture in its ability to quickly latch onto strong sentiment signals within the reviews. As expected, the validation loss stabilized around epoch 5, but then sharply rose, indicating overfitting quickly after the first few epochs.
https://wandb.ai/jbsmith-hamilton-college/CS-366%20Assignment%203



Final evaluation results
Test Accuracy: 0.87667

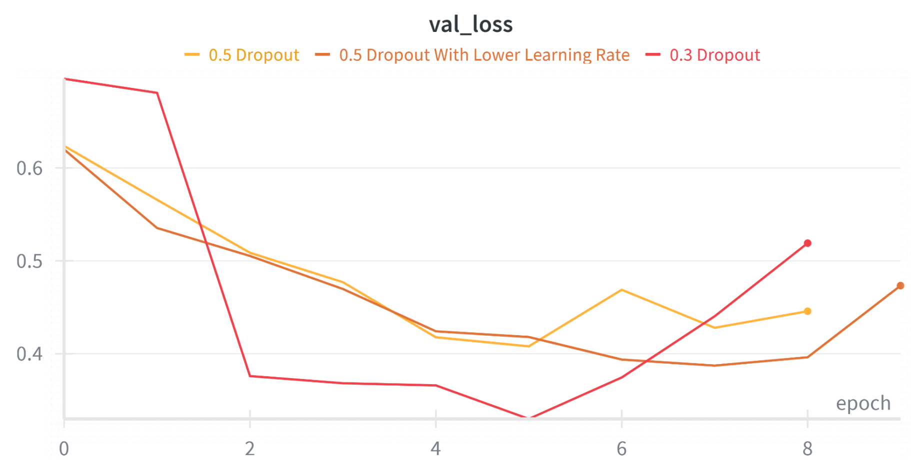


Figure 1. Validation loss graphs for different learning rate/dropout combinations. As shown, the 0.5 dropout runs didn't achieve as low levels of validation loss as the run with 0.3 dropout

Confusion Matrix:	Predicted Negative (0)	Predicted Positive (1)
Actual Negative (0) |	3328	407
Actual Positive (1) |	519	3246

Classification report

	Precision	Recall	F1-score	Support
negative	0.8651	0.8910	0.8779	3735
positive	0.8886	0.8622	0.8752	3765
accuracy			0.8765	7500
macro avg	0.8768	0.8766	0.8765	7500
weighted avg	0.8769	0.8765	0.8765	7500

3 Misclassified Examples:

Negative misclassified as positive

{"text": "Sorry, folks, but all of you that say this is a great documentary... and that award it won at Sundance... well, you've all been duped. I've heard for a few years how I had to see this documentary and I finally watched it. Maybe in 1999, when it came out, and reality TV didn't have such a dominant presence in the industry, this movie would have seemed entertaining. But Mike and Mark are so obviously playing themselves, Mike and Mark. At times they are funny and some of the lines seem off the cuff, but mostly they do not ring true. They are the reality version of Jay and Silent Bob. Yes the people are real, they are not actors, but it's put on, it's an exaggeration of themselves, and it's so obvious that it's hard to believe so many people think it's the real deal. I wasn't fooled so it was actually a tad boring. Mildly amusing, but not missing much if you miss it.", "true": 0, "pred": 1}

{"text": "Rigoletto is Verdi's masterpiece, full of drama, emotion and powerful, memorable music. The maestro must have rolled in his grave when this bawdy travesty of his work was released with its needless frontal nudity and cheap copulating and its portrayal of the naive but principalled Gilda as a horny ditz. Opera certainly can be adapted to cinema --- look at Zeferelli's magnificent La Traviata --- but when a work is as superb as Rigoletto, it doesn't need cheap gimmicks. It might even have been acceptable if the dubbed in music had been good but it is a mediocre rendering of the libretto with second rate sound quality at that.", "true": 0, "pred": 1}

Positive misclassified as negative

{"text": "I don't think the world was ready for this film. I know I wasn't. I'd been expected a standard low-budget schlock exploitation potboiler. Instead, I got the most intelligent reworking of Shakespeare since Peter Greenaway's "Prospero's Books". This should become the definitive film version of Romeo And Juliet. It won't of course. But that's the world's loss.", "true": 1, "pred": 0}

Discussion and conclusion

Looking at the misclassification examples, it’s understandable how these specific reviews were misclassified since negative reviews also contain lots of positive language, like “great documentary,” which could be what mixed up the model. With a final test accuracy of 88%, overall the model performed pretty well. We learned that Bi-LSTM seemed to work specifically for this task. A unidirectional RNN would likely have missed these dependencies, whereas our Bi-LSTM mostly captured the context from both forward and backward passes. As discussed in class though, using a transformer would probably help achieve a higher accuracy than both. In conclusion, we learned how to implement A Bi-LSTM architecture and achieved positive results for the IMDB movie review classification task.

References

https://www.sabrepc.com/blog/Deep-Learning-and-AI/text-classification-with-bert
Explains how to initialize a generic text classification dataset

https://lightning.ai/docs/pytorch/LTS/common/lightning_module.html
Used to help implement PyTorch Lightning in code